Two-Minute Interview Narration

A concise explanation of the full multi-stage NLP solution.

Project: Automated Mobile-Technology Brand Sentiment and Headline Generator

Source archive: NLP-Sentiment-Classification-of-Mobile-brands-main.zip | Notebook: main.ipynb | Supporting report and presentation included

Narration (approximately two minutes)

“I worked on an automated headline and brand-sentiment generation project for multilingual articles and tweets. The business objective was to identify mobile-technology content, find the mobile brands mentioned, understand sentiment toward each brand separately, and generate a short English headline for relevant articles. The input could be English, Hindi or Hinglish.”

“We designed it as a cascade of tasks. First, we clean the text by removing URLs, redundant symbols and tweet-specific markers, while converting emojis into text descriptors so sentiment information is retained. We then use TF-IDF features with an XGBoost classifier to identify whether an article or tweet is about mobile technology. This stage processes the largest volume, so a lightweight model is preferred.”

“For mobile-tech content, we identify brands using regular-expression matching against a curated dictionary of roughly 150 brands. We chose this over a general NER model because the brand vocabulary is bounded and regex is fast and easy to update. We detect the language and translate Hindi and Hinglish content into English. Then we isolate the context associated with each brand. For sentiment, the project explores contextual BERT-based classification, dependency and lexicon approaches, and lightweight TF-IDF classifiers for single-brand cases. An ensemble combines the approaches, with long articles processed in chunks.”

“For mobile-tech articles, we fine-tuned T5-base on translated English text to generate abstractive headlines. The report says T5 was trained for two epochs with validation-based early stopping and was preferred over PEGASUS and a Lead-3 baseline.”

“In the supplied evaluation notebook, the sample run processed 222 articles and 210 tweets initially, predicted 71 articles and 62 tweets as mobile-tech, and produced 557 brand-level output rows. Total recorded runtime was about 362 seconds, with translation taking about 290 seconds, making it the main bottleneck. The archive does not provide final numeric task metrics or all training assets, so I would not claim an overall score. My next improvements would be reliable translation, better brand-context resolution, factuality checks for generated headlines, and complete per-language evaluation.”

Likely interview questions

1. Why is this a pipeline instead of one model? Different tasks have different outputs and costs; cascading filters save expensive inference.

2. Why TF-IDF + XGBoost for theme classification? It is compact, fast and effective for a binary text classification filter.

3. Why regex instead of NER? The known brand list is small and curated; regex is fast and deterministic, though weaker for unseen aliases and ambiguity.

4. Why translate into English? Stronger English pretrained models and headline generation were available; translation improved performance in their experiments but adds latency and error propagation.

5. How is brand-specific sentiment different from document sentiment? A text can praise one brand and criticize another, so the model must condition sentiment on the target brand and its context.

6. Why use BERT? Contextual embeddings capture word meaning in context and can distinguish sentiment around different brand mentions better than simple bag-of-words features.

7. Why use a lightweight model for single-brand text? It reduces latency when the target context is simpler; larger contextual models are reserved for ambiguous or multi-brand cases.

8. Why T5? It is an encoder-decoder text-to-text model suited to abstractive generation; the report says it outperformed PEGASUS and Lead-3 on their validation metric.

9. What was the main bottleneck? Translation, which took about 290 seconds out of a 362-second sample run.

10. What is the biggest technical risk? Error propagation across language detection, translation, regex entity extraction and brand-context segmentation.

11. How would you evaluate it? Theme F1, brand entity accuracy, brand-specific sentiment F1, headline ROUGE/BLEU/embedding similarity, plus language-sliced metrics and latency.

12. What is the limitation of the supplied repo? It lacks the original datasets, model files and source modules required to reproduce the full training/inference pipeline independently.